Concept of variation

1 Suppose the constant of variation, k, is positive.

a If y varies directly with x, describe how y changes as x increases and decreases.

b If y varies inversely with x, describe how y changes as x increases and decreases.

2 State whether the following equations represent direct or inverse variation:

a $y = 5x$

b $y = \frac{5}{x}$

3 The period of a pendulum varies directly with the square root of its length. If the length is quadrupled, what happens to the period?

4 State whether the following are examples of direct or inverse variation:

a The variation relating the distance between two locations on a map and the actual distance between the two locations.

b The variation relating the number of workers hired to build a house and the time required to build the house.

c Rhe variation relating the time it takes an ice cube to melt in water and the temperature of the water.

Direct variation

5 In the equation $y = 3x$, y varies directly as x.

a Find the value of y when $x = 10$.

b Find the value of y when $x = 5$.

c For these two ordered pairs, what is the result when the y value is divided by the x value?

6 Find the equation relating p and q given the table of values:

p	3	6	9	12
q	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$

7 Find the equation relating t and s given the table of values:

s	1	2	3	4

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8 If $d = \frac{1}{5}s$, where d is the approximate distance (in miles) from a storm, and s is the number of seconds between seeing lightning and hearing thunder, describe the relationship between the distance and the number of seconds.

9 If $r = \frac{d}{t}$, where r is the speed when d kilometres in t hours, describe the relationship between the speed and the kilometres.

10 If $f = \frac{mv^2}{r}$, where f is the centripetal force of an object of mass m moving along a circle of radius r at velocity v, describe the relationship between the centripetal force and the mass.

11 The mass in grams, M, of a cube of cork varies directly with the cube of the side length in centimetres, x. If a cubic centimetre of cork has a mass of 0.29:

- **a** Find the constant of variation, k.
- **b** Express M in terms of x.
- **c** Find the mass of a cube of cork with a side length of 8 centimetres correct to two decimal places.

12 The surface area, A, of a regular tetrahedron varies directly with the square of its side length, s. A particular tetrahedron with a side length of 2 cm has a surface area of 6.93.

- **a** Find the constant of variation, k, to two decimal places.
- **b** Using the rounded value of k, express A in terms of s.
- **c** Find the surface area of a tetrahedron with a side length of 3 cm to two decimal places.

13 The area, A, of an equilateral triangle varies directly with the square of its side length, s. An equilateral triangle with a side length of 7 cm has an area of 21.22.

- **a** Find the constant of variation, k, to two decimal places.
- **b** Express A in terms of s.
- **c** Find the area of an equilateral triangle with a side length of 2 cm to two decimal places.

14 The number of eggs, n, used in a recipe for a particular cake varies directly with the square of the diameter of the tin, d, for tins with constant depth. 2 eggs are used in a recipe for a tin with a diameter of 17 cm.

- **a** Find the exact value of the constant of variation, k.
- **b** How many eggs, n, would be used for a tin with a diameter of 39 cm?

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Inverse variation

15 State the meaning of inverse proportion.

16 If r varies inversely with a, write an equation that uses k as the constant of variation.

17 In the equation $y = \frac{18}{x}$, y varies inversely with x. When $x = 6$, $y = 3$.

a Solve for y when $x = 2$.

b For these two ordered pairs, what is the result when the y -value is multiplied by the x -value?

18 Suppose that y varies inversely as the square of x, and that $y = 0.0023$ when $x = 0.4$.

a Find the constant of variation, k. Round your answer correct to six decimal places.

b Write the variation equation for y in terms of x.

19 State whether the following equations are examples of inverse variation:

a $y = \frac{7}{x}$

b $y = 6x + 8$

c $y = -\frac{9}{x}$

d $y = \frac{8}{x^2}$

e $y = 2x^2 - 7x - 4$

f $y = 3 - x$

g $x = 1 + y^3$

h $x = \frac{8}{y^2}$

i $y = 6x + 8$

j $xy = -7$

k $x = \frac{2}{y}$

l $xy = 5x$

20 Consider the inverse variation equation $y = \frac{6}{x}$.

a Complete the following tables:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

b Plot the data from the table of values on a number plane.

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21 If y is inversely proportional to x, and $y = 20$ when $x = 10$:

a Find the constant of variation, k.

b Express y in terms of x.

c Find the value of y when $x = 5$.

22 Which of the following tables could represent an inversely proportional relationship between x and y?

A

x	1	2	3	4
y	3	1.5	1	0.75

B

x	1	2	3	4
y	36	18	12	9

C

x	1	5	6	10
y	3	75	108	300

D

x	1	2	3	4
y	4	5	6	7

23 Find the equation relating n and r given the table of values:

n	1	2	3	4
r	5	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$

24 Consider the equation $s = \dfrac{375}{t}$.

a State the constant of proportionality.

b Find the value of s when $t = 6$.

c Find the value of s when $t = 12$.

25 The time, t, taken by a typist to type up a document is inversely proportional to his typing speed, s. That is, the quicker the typing speed, the less time it will take. If it takes a typist 20 minutes to type a particular document, typing at a speed of 61 words per minute:

a Find the constant of variation k.

b How long (in minutes) will it take a typist with a typing speed of 30.5 words per minute to type up the document?

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